

CerviXAI

Complete Android App and FastAPI Backend Run Guide

Environment covered: Windows Command Prompt, physical Android phone, USB debugging, local FastAPI backend, and ADB reverse port forwarding.

Component	Exact location
Android project	C:\Users\user\AndroidStudioProjects\CervicalCancerXAI
Backend project	D:\Cervical_Cancer_XAI
Android Studio JDK	C:\Program Files\Android\Android Studio\jbr
Android SDK	C:\Users\user\AppData\Local\Android\Sdk
ADB executable	C:\Users\user\AppData\Local\Android\Sdk\platform-tools\adb.exe
Backend virtual environment	D:\Cervical_Cancer_XAI\venv

Important: All commands in this guide are for Windows Command Prompt (CMD), not PowerShell.

1. How the local setup works

- FastAPI runs on the computer at port 8000.
- The Android app is installed on the phone and opens normally from its launcher icon.
- ADB reverse maps the phone address 127.0.0.1:8000 to the computer backend through USB.
- The phone may keep cellular data enabled, but CerviXAI backend traffic uses the USB connection.
- The backend is required for predictions. Login and locally stored account data can work without it.

Required Android API URL

For the USB plus ADB reverse setup, ApiConfig.kt must contain:

```
package com.ranjitha.cervicalcancerxai.api

object ApiConfig {
    const val BASE_URL = "http://127.0.0.1:8000/"
}
```

Do not use your-backend.example.com. That is only a placeholder and causes Unable to resolve host.

2. First-time backend setup

Open Command Prompt and run each line separately:

```
cd /d D:\Cervical_Cancer_XAI
python -m venv .venv
.venv\Scripts\activate
python -m pip install --default-timeout=120 --retries 10 -r requirements.txt
```

When activated, the prompt begins with (.venv). Package installation is normally needed only once.

If PyPI DNS or timeout errors appear

```
nslookup pypi.org
nslookup files.pythonhosted.org
ipconfig /flushdns
python -m pip install --index-url https://pypi.org/simple --trusted-host pypi.org --trusted-host
files.pythonhosted.org --default-timeout=120 --retries 10 -r requirements.txt
```

If DNS lookup still fails, retry on a different network such as a phone hotspot.

3. Start the backend each day

```
cd /d D:\Cervical_Cancer_XAI
.venv\Scripts\activate
python -m uvicorn src.api:app --host 0.0.0.0 --port 8000
```

Expected result:

```
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000
```

Keep this Command Prompt window open. Press Ctrl+C only when you want to stop the backend.

Test the backend on the computer

```
curl http://127.0.0.1:8000/
curl "http://127.0.0.1:8000/cancer-report?patient_id=0"
```

Interactive API documentation is available at <http://127.0.0.1:8000/docs> in the computer browser.

4. First-time Android build and installation

Open another Command Prompt:

```
cd /d C:\Users\User\AndroidStudioProjects\CervicalCancerXAI
set "JAVA_HOME=C:\Program Files\Android\Android Studio\jbr"
set "PATH=%JAVA_HOME%\bin;%PATH%"
gradlew.bat assembleDebug
```

The generated APK is located at:

```
C:\Users\user\AndroidStudioProjects\CervicalCancerXAI\app\build\outputs\apk\debug\app-debug.apk
```

Connect and authorize the phone

```
set "ADB=C:\Users\user\AppData\Local\Android\Sdk\platform-tools\adb.exe"
"%ADB%" devices
```

The phone must be unlocked. Accept Allow USB debugging and choose Always allow from this computer. The device list must show device, not unauthorized or offline.

Install the app

```
cd /d C:\Users\User\AndroidStudioProjects\CervicalCancerXAI
gradlew.bat installDebug
```

Expected result: Installed on 1 device and BUILD SUCCESSFUL.

5. Daily start procedure - exact order

Window 1 - Backend

```
cd /d D:\Cervical_Cancer_XAI
.venv\Scripts\activate
python -m uvicorn src.api:app --host 0.0.0.0 --port 8000
```

Window 2 - Phone connection

```
set "ADB=C:\Users\user\AppData\Local\Android\Sdk\platform-tools\adb.exe"
"%ADB%" devices
"%ADB%" reverse tcp:8000 tcp:8000
```

A successful reverse command prints 8000. Re-run it after reconnecting the cable or restarting ADB.

Open the app

Open CerviXAI normally from the phone launcher. No frontend build, installation, or activity command is needed every day.

Optional launch command:

```
"%ADB%" shell am start -n com.ranjitha.cervicalcancerxai/.MainActivity
```

6. After changing Android code

Only rebuild and reinstall when source code, resources, dependencies, or the API URL changes.

```
cd /d C:\Users\user\AndroidStudioProjects\CervicalCancerXAI
set "JAVA_HOME=C:\Program Files\Android\Android Studio\jbr"
gradlew.bat installDebug
set "ADB=C:\Users\user\AppData\Local\Android\Sdk\platform-tools\adb.exe"
"%ADB%" reverse tcp:8000 tcp:8000
"%ADB%" shell am force-stop com.ranjitha.cervicalcancerxai
"%ADB%" shell am start -n com.ranjitha.cervicalcancerxai/.MainActivity
```

7. Build checks

```
cd /d C:\Users\user\AndroidStudioProjects\CervicalCancerXAI
set "JAVA_HOME=C:\Program Files\Android\Android Studio\jbr"
gradlew.bat testDebugUnitTest
gradlew.bat lintDebug
gradlew.bat assembleDebug
```

Combined validation command:

```
gradlew.bat testDebugUnitTest lintDebug assembleDebug
```

8. Troubleshooting

JAVA_HOME is not set

```
set "JAVA_HOME=C:\Program Files\Android\Android Studio\jbr"
set "PATH=%JAVA_HOME%\bin;%PATH%"
java -version
```

adb is not recognized

```
set "ADB=C:\Users\user\AppData\Local\Android\Sdk\platform-tools\adb.exe"
"%ADB%" devices
```

Device is unauthorized

- Unlock the phone.
- Enable Developer options and USB debugging.
- Reconnect the cable and select File transfer mode.
- Accept the RSA authorization prompt.

- If necessary, choose Revoke USB debugging authorizations on the phone and reconnect.

Device is offline

```
"%ADB%" kill-server
"%ADB%" start-server
"%ADB%" devices
"%ADB%" reverse tcp:8000 tcp:8000
```

Prediction failed: Unable to resolve host

Confirm ApiConfig.kt uses `http://127.0.0.1:8000/` for this USB setup. Rebuild and reinstall after changing it.

Prediction failed: connection refused or timeout

```
curl http://127.0.0.1:8000/
"%ADB%" reverse --list
"%ADB%" reverse tcp:8000 tcp:8000
```

Also confirm the backend window is still running and the USB device status is device.

Start command was accidentally joined to another command

Run one command per line. The correct names are `JAVA_HOME` and `gradlew.bat`. Do not type `JAVA\_HOME` or `gradlew\.bat`.

9. Accounts, password reset, and history

- Accounts are currently stored locally on the Android phone, not in FastAPI.
- Login accepts the registered username or email after the LoginScreen update.
- Clearing app data deletes the locally registered account and password.
- Prediction History is currently a placeholder and does not save API results.

Clear all local app data only when intentionally creating a fresh account:

```
"%ADB%" shell pm clear com.ranjitha.cervicalcancerxai
```

10. Quick daily checklist

Step	Check
1	Start FastAPI and keep its window open.
2	Connect and unlock the phone.
3	Confirm adb devices shows device.
4	Run adb reverse tcp:8000 tcp:8000.
5	Open CerviXAI from the phone icon.
6	Run an assessment and confirm the backend logs GET /cancer-report ... 200 OK.

Normal daily use does not require rebuilding or reinstalling the Android frontend. Start the backend, establish ADB reverse, and open the installed app.